

Module 6: Organic chemistry and analysis

The content within this module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry and Module 4: Core organic chemistry.

This module introduces several new functional groups and emphasises the importance of organic synthesis. This module also adds NMR spectroscopy to the instrumentation techniques used in organic and forensic analysis.

The main areas of organic chemistry studied include:

- aromatic compounds
- carboxylic acids and esters
- organic nitrogen compounds: amines and amino acids
- polymerisation: addition polymers and condensation polymers
- synthetic organic chemistry and further development of practical skills
- the importance of modern analytical techniques in organic analysis.

Synoptic assessment

This module provides a context for synoptic assessment and the subject content links strongly with the content encountered in Module 2: Foundations in chemistry and Module 4: Core organic chemistry.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure
- Organic nomenclature and structures
- Hydrocarbons
- Alcohols and haloalkanes
- Synthesis and analysis

Knowledge and understanding of Module 2 and Module 4 will be assumed and examination questions will be set that link their content with this module and other areas of chemistry.

6.1 Aromatic compounds, carbonyls and acids

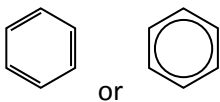
This section extends the range of functional groups encountered in Module 4.

Aromatic compounds are first introduced, including the central role of delocalisation within the chemistry of arenes and phenols. Directing groups are also introduced, including their importance to organic synthesis.

The important carbonyl compounds, aldehydes and ketones, are then studied.

Finally, carboxylic acids and their related functional groups, acyl chlorides and esters, are studied. The importance of acyl chlorides in organic synthesis is emphasised.

6.1.1 Aromatic compounds

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Benzene and aromatic compounds	
(a) the comparison of the Kekulé model of benzene with the subsequent delocalised models for benzene in terms of p-orbital overlap forming a delocalised π -system	Learners may represent the structure of benzene in equations and mechanisms as:  HSW1,7 Development of the model for benzene over time.
(b) the experimental evidence for a delocalised, rather than Kekulé, model for benzene in terms of bond lengths, enthalpy change of hydrogenation and resistance to reaction (see also 6.1.1 f)	HSW11 Acceptance of the delocalised benzene model by the scientific community in light of supporting experimental evidence.
(c) use of IUPAC rules of nomenclature for systematically naming substituted aromatic compounds	Use of locant numbers to identify positions of substitution e.g. 2,4-dinitromethylbenzene. HSW8 Introduction of systematic nomenclature.
Electrophilic substitution	
(d) the electrophilic substitution of aromatic compounds with: (i) concentrated nitric acid in the presence of concentrated sulfuric acid (ii) a halogen in the presence of a halogen carrier (iii) a haloalkane or acyl chloride in the presence of a halogen carrier (Friedel–Crafts reaction) and its importance to synthesis by formation of a C–C bond to an aromatic ring (see also 6.2.4 d)	Halogen carriers include iron, iron halides and aluminium halides.
(e) the mechanism of electrophilic substitution in arenes for nitration and halogenation (see also 4.1.1 h–i)	For nitration mechanism, learners should include equations for formation of NO_2^+ . Halogen carriers include iron, iron halides and aluminium halides. For the halogenation mechanism, the electrophile can be assumed to be X^+ . HSW1,2,8 Use of reaction mechanisms to explain organic reactions.

- (f) the explanation of the relative resistance to bromination of benzene, compared with alkenes, in terms of the delocalised electron density of the π -system in benzene compared with the localised electron density of the π -bond in alkenes (**see also 4.1.3 a, 6.1.1 a**)
- (g) the interpretation of unfamiliar electrophilic substitution reactions of aromatic compounds, including prediction of mechanisms

HSW2,5 Use of delocalised benzene model to explain reactivity.

Extra information may be provided on exam papers.

Phenols

- (h) the weak acidity of phenols shown by the neutralisation reaction with NaOH but absence of reaction with carbonates (**see also 5.1.3 b**)
- (i) the electrophilic substitution reactions of phenol:
- (i) with bromine to form 2,4,6-tribromophenol
 - (ii) with dilute nitric acid to form a mixture of 2-nitrophenol and 4-nitrophenol
- (j) the relative ease of electrophilic substitution of phenol compared with benzene, in terms of electron pair donation to the π -system from an oxygen p-orbital in phenol (**see also 4.1.3 a**)
- (k) the 2- and 4-directing effect of electron-donating groups (OH, NH_2) and the 3-directing effect of electron-withdrawing groups (NO_2) in electrophilic substitution of aromatic compounds
- (l) the prediction of substitution products of aromatic compounds by directing effects and the importance to organic synthesis (**see also 6.2.5 Organic Synthesis**).

PAG7 (see also 6.3.1 c)

Note that nitration with phenol does not require concentrated HNO_3 or the presence of a concentrated H_2SO_4 catalyst.

Illustrated by reactions with bromine and with nitric acid.
Explanation is only in terms of susceptibility of ring to 'attack' and **not** in terms of stability of intermediate.

HSW2,5 Use of delocalised benzene model to explain reactivity.

Learners will **not** be expected to know further electron-donating or electron-withdrawing groups; relevant additional data will be supplied in examinations.

HSW5 Correlation between substituted group and position of reaction.